Circle Packing via Soft Actor-Critic and Bound-Constrained Optimization

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Abstract

Circle packing is a classical NP-hard problem characterized by a non-convex energy landscape and a multitude of jamming states, which poses significant challenges for traditional numerical optimization and heuristic methods. While recent advancements in Deep Reinforcement Learning (DRL) have shown promise in combinatorial optimization, generic RL approaches often fail to satisfy the high-precision geometric constraints required for dense packing. This study proposes a hierarchical "Training-Sampling-Polishing" architecture. By establishing a custom differentiable physics environment and leveraging the maximum entropy property of the Soft Actor-Critic (SAC) algorithm, we achieve robust global topological exploration. Subsequently, we integrate the **L-BFGS-B** (Limited-memory Broyden–Fletcher–Goldfarb–Shanno with Bound constraints) algorithm to perform deterministic geometric refinement. Experimental results on a 26-circle instance yield a radius sum of 2.62, demonstrating that this neuro-symbolic approach effectively reconciles the conflict between global search capabilities and sub-pixel solution precision.

Keywords: Circle Packing, Deep Reinforcement Learning, Soft Actor-Critic, L-BFGS-B, Neuro-Symbolic AI, Differentiable Physics

1 Introduction

Traditional approaches to circle packing have predominantly relied on numerical optimization and physics-based simulations [4, 9]. While potential energy descent methods ensure convergence to local optima, the inherent non-convexity of the problem renders them highly susceptible to 'jamming states,' making the discovery of the global optimal topology challenging [9]. In recent years, Deep Reinforcement Learning (DRL) has achieved significant breakthroughs in combinatorial optimization [2]. Particularly in 2D layout and packing problems, RL has demonstrated search capabilities superior to those of traditional heuristic algorithms [6]. Notably, the developers of the OpenEvolve framework utilized circle packing as a demonstrative case study, achieving impressive results by leveraging the iterative trial-and-error and feedback mechanism of the Large Language Model, Gemini. However, generic RL methods often struggle to satisfy high-precision geometric constraints. Inspired by Google's work on chip placement [7], this study proposes a neuro-symbolic hybrid architecture. By leveraging the maximum entropy property of Soft Actor-Critic (SAC) for global topological exploration and integrating L-BFGS-B for deterministic geometric refinement, our approach simultaneously achieves robust global search capabilities and sub-pixel solution precision.

2 Problem Analysis and Physics Environment

The circle packing problem is inherently non-convex and NP-hard. It is characterized by a rugged energy landscape populated with an exponential number of local minima. Each local minimum corresponds to a mechanically stable state defined by a unique rigid contact network. Standard gradient-based optimization methods are prone to getting trapped in these basins of attraction, as transitioning to a better configuration often requires a global topological rearrangement. For example, if the triangulation formed by the circle centers is rigid, the agent cannot reach other stable configurations by moving a single circle. Reaching alternative configurations requires traversing lower-scoring states or causing temporary overlaps between circles.

3 Methodology: The Training-Sampling-Polishing Architecture

We proposed a hierarchical "Training-Sampling-Polishing" architecture, a neuro-symbolic framework designed to reconcile the conflict between global exploration and local precision in the NP-hard circle packing problem. The methodology decomposes the optimization process into two distinct phases: Stochastic Topology Search with low penalty and Deterministic Geometric Refinement (driven by Numerical Optimization).

3.1 Phase I: Global Topology Search via Soft Actor-Critic (SAC)

The initial phase employs the SAC agent to interact with a custom differentiable physics environment. SAC (Soft Actor-Critic) is a state-of-the-art, model-free, off-policy Deep Reinforcement Learning algorithm designed for continuous control tasks [5]. It is currently considered the "Gold Standard" for problems involving complex physics and continuous action spaces. Standard RL algorithms (like DDPG or PPO) aim to maximize only the expected cumulative reward:

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} [r(\mathbf{s}_t, \mathbf{a}_t)] \tag{1}$$

In contrast, SAC aims to maximize the reward plus the entropy (randomness) of the policy. The objective function is augmented as follows:

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} [r(\mathbf{s}_t, \mathbf{a}_t) + \alpha \mathcal{H}(\pi(\cdot|\mathbf{s}_t))] \tag{2}$$

where $\mathcal{H}(\pi(\cdot|\mathbf{s}_t))$ represents the entropy of the policy at state $\mathbf{s}_t$, and α is the temperature parameter determining the relative importance of the entropy term. By maximizing entropy, the agent is encouraged to explore multiple ways to achieve the goal rather than converging to a single, deterministic behavior too quickly. Driven by the Maximum Entropy principle, the SAC agent learns to navigate the rugged, non-convex energy landscape of the packing problem. Its primary objective is not to immediately resolve sub-pixel collisions, but to overcome "jamming transitions"—the local minima where circles are mechanically interlocked in suboptimal configurations, and explores diverse topological arrangements. In the training process, distinct from conventional 'black-box' rigid body simulations, this research establishes a custom differentiable physics environment specifically tailored for the NP-hard circle packing problem. The core innovation lies in the formulation of a potential-based quasi-static simulation that functions as a dynamic solver for Laguerre-Voronoi (Power) Diagrams.

3.1.1 Geometric Definitions: Laguerre-Voronoi Diagram

Let $\mathcal{C} = \{C_1, C_2, \ldots, C_n\}$ denote a set of circles in a 2D planar domain $\Omega \subset \mathbb{R}^2$, where each circle C_i is defined by its center $\mathbf{c}_i \in \mathbb{R}^2$ and radius $r_i \in \mathbb{R}^+$. Unlike standard Voronoi diagrams based on Euclidean distance, circle packing with unequal radii requires the introduction of the *Power Distance*, which represents the squared tangential distance from a point to a circle [1]:

$$d_{\text{pow}}(\mathbf{x}, C_i) = \|\mathbf{x} - \mathbf{c}_i\|^2 - r_i^2 \tag{3}$$

where $\|\cdot\|$ denotes the Euclidean norm. Based on this metric, the *Laguerre-Voronoi Cell* (or Power Cell) V_i associated with circle C_i is defined as the region where C_i is the closest circle in terms of power distance [1]:

$$V_i = \{\mathbf{x} \in \Omega \mid d_{\text{pow}}(\mathbf{x}, C_i) \leq d_{\text{pow}}(\mathbf{x}, C_j), \forall j \neq i\} \tag{4}$$

For any two adjacent circles C_i and C_j, the geometric constraint that a point $\mathbf{x}$ belongs to cell V_i rather than V_j is defined by the following inequality (which forms the half-plane boundary):

$$\|\mathbf{x} - \mathbf{c}_i\|^2 - r_i^2 \leq \|\mathbf{x} - \mathbf{c}_j\|^2 - r_j^2 \tag{5}$$

A valid, non-overlapping packing configuration must satisfy the geometric constraint:

$$\forall i \in \{1, \ldots, n\}, \quad C_i \subset V_i \tag{6}$$

This implies that each circle must be strictly contained within its own Laguerre cell. When two circles are tangent, the contact point $\mathbf{p}_{ij}$ must lie precisely on the radical axis.

3.1.2 Physics Engine: Soft Potential Field

To solve the aforementioned geometric constraints in a continuous space, we construct a differentiable *Soft Potential Field.* Let the Euclidean distance between the centers of circle i and circle j be $d_{ij} = \|\mathbf{c}_i - \mathbf{c}_j\|$. We define the *Overlap Magnitude* δ_{ij}, which corresponds to the penetration depth commonly used in computational mechanics [9], as:

$$\delta_{ij}(\mathbf{c}_i, \mathbf{c}_j, r_i, r_j) = \max(0, (r_i + r_j) - d_{ij}) \tag{7}$$

To obtain a smooth and locally convex gradient, the total potential energy function $U(\mathbf{s})$ of the physics engine is defined as the sum of squared overlaps (corresponding to Hookean elastic potential energy). Let the state vector be $\mathbf{s} = [\mathbf{c}_1, \ldots, \mathbf{c}_n, r_1, \ldots, r_n]^T$. The potential energy is:

$$U(\mathbf{s}) = \sum_{i=1}^{n} \sum_{j=i+1}^{n} (\delta_{ij}(\mathbf{c}_i, \mathbf{c}_j, r_i, r_j))^2 \tag{8}$$

3.1.3 Quasi-static Dynamics & Optimization

The evolution of the physics engine follows *Quasi-static Dynamics*, ignoring inertial terms and considering only displacements driven by the potential gradient. In the *Polishing Phase*, the problem is formulated as a bound-constrained non-linear programming problem.

$$\begin{aligned} \min_{\mathbf{s}} \quad & U(\mathbf{s}) \\ \text{s.t.} \quad & \mathbf{c}_i \in [r_i, 1 - r_i]^2, \quad i = 1, \ldots, n \end{aligned} \tag{9}$$

We employ the **L-BFGS-B** algorithm to solve this optimization. Its update rule utilizes an approximation of the Hessian matrix of the potential field $\mathbf{B} \approx \nabla^2_{\mathbf{s}} U(\mathbf{s})$. According to the quasi-Newton condition, $\mathbf{B}_k$ is updated iteratively to satisfy the secant equation based on curvature information [8]:

$$\mathbf{s}_{k+1} = \mathbf{s}_k - \gamma_k \mathbf{B}_k^{-1} \nabla_{\mathbf{s}} U(\mathbf{s}_k) \tag{10}$$

Since $U(\mathbf{s})$ is convex in the local region where $\delta_{ij} > 0$, second-order optimization algorithms can drive the system toward the state $U(\mathbf{s}) \to 0$ with super-linear convergence rates.

3.1.4 Connection: Duality

When the system converges to $U(\mathbf{s}) = 0$ and the packing density is maximized, the set of circle centers $\{\mathbf{c}_i\}$ and the contact network $G = (V, E)$ form a *Weighted Delaunay Triangulation*, which is the dual graph of the Laguerre-Voronoi diagram. The gradient flow ∇U in the physics engine is geometrically equivalent to driving the centers $\mathbf{c}_i$ along the direction of steepest descent to move away from the radical axis and retreat inside their respective Laguerre cells V_i. Thus, our physical simulation is essentially a dynamic process of constructing an optimal Laguerre-Voronoi tessellation.

3.2 Phase II: Sampling

During the Sampling stage, the trained policy functions as a generative heuristic, producing a distribution of candidate layouts. These candidates represent coarse-grained solutions where the relative positions (the weighted Delaunay triangulation) are fundamentally correct, but the geometric parameters remain relaxed.

3.3 Phase III: Deterministic Geometric Refinement via L-BFGS-B

The final phase transitions from stochastic exploration to deterministic convergence using the L-BFGS-B algorithm. This step transforms the "soft" topological proposal provided by the neural network into a "hard" mathematical proof of the packing configuration. L-BFGS-B is a sophisticated Quasi-Newton optimization method selected for its efficacy in handling high-dimensional, bound-constrained problems [3]. Its superiority over first-order methods (such as SGD or Adam) emerges from its ability to utilize second-order information:

1. **Curvature Approximation (The Hessian):** Unlike algorithms that rely solely on the gradient (slope), L-BFGS-B approximates the Hessian matrix (curvature)

using a limited history of past gradient updates. This allows the solver to model the local energy landscape at iteration k as a quadratic function $m_k(\mathbf{x})$:

$$m_k(\mathbf{x}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T(\mathbf{x} - \mathbf{x}_k) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_k)^T \mathbf{B}_k (\mathbf{x} - \mathbf{x}_k) \tag{11}$$

where $\mathbf{B}_k$ is the approximate Hessian. This enables it to take "Newton steps" that cut directly towards the valley floor rather than oscillating across it.

2. **Bound Constraint Handling (The 'B'):** The algorithm utilizes the Gradient Projection Method to rigorously enforce the geometric constraints (e.g., box boundaries $0 \leq x \leq 1$). It identifies the "active set" of variables lying on the boundaries and optimizes the remaining free variables within a "reduced subspace," ensuring that the physical constraints are respected with analytical precision.

So in this system, L-BFGS-B exploits the local convexity of the potential energy function within the basin of attraction identified by the SAC agent, achieving superlinear convergence rates, rapidly minimizing the overlap penalty to machine precision (tolerance 10^{-9}) and maximizing the packing density. This ensures that the final output satisfies the strict tangency conditions required by the mathematical definition of a valid packing.

Additionally, we implemented several auxiliary strategies to further enhance model performance. These include:

1. **Stepwise Lambda Scheduling:** During training, we employed a stepwise schedule for the penalty coefficient (λ). This ensures the model prioritizes learning superior topological structures first, preventing premature convergence (or 'jamming') caused by an excessive aversion to overlap penalties in the early stages.

2. **Adaptive Radius Shrinking:** During the deterministic refinement phase, if significant overlaps persist, the algorithm is permitted to iteratively reduce the circle radii. This relaxation mechanism allows the solver to escape local optima and explore a broader search space for feasible solutions.

4 Comparative Study: Graph Neural Networks

As a comparison, we also explored the use of Graph Neural Network (GNN) algorithms to train models for this problem. First, the core mechanism of GNNs is highly compatible with the physical nature of the circle packing problem. Circle packing is essentially a constrained local interaction problem: within a unit square, the radius of any given circle is primarily restricted by the "squeezing" effect of its immediate neighbors. By treating the center coordinates (x, y) as node features, the unique message-passing mechanism of GNNs can effectively capture the interference relationships between adjacent circles through edge connections. Furthermore, GNNs possess permutation invariance, meaning that the geometric logic dictated by physical positions remains consistent regardless of how the circles are indexed. To enable a general GNN to handle high-precision geometric constraints, I implemented several targeted architectural refinements:

- During the graph building phase, static graphs were replaced with dynamic K-Nearest Neighbors (KNN) graphs. This simulates the physical contact between a circle and its approximately six neighbors, ensuring the model focuses on the most direct constraints.

- For the core operator, GATConv (Graph Attention Network) was introduced, allowing the model to automatically identify and assign higher weights to closer, more "threatening" neighbors.
- In the output layer, to satisfy the physical requirement that radii must be strictly positive, a Softplus activation function was utilized. This provides a smoother gradient space compared to ReLU.

However, GNNs still exhibit certain limitations when addressing such "hard-constraint" optimization problems. As statistical learners, it is difficult for GNNs to spontaneously generate perfect predictions that are entirely overlap-free and boundary-compliant at the 10^{-8} error level required for this task. Additionally, the solution space is fraught with extremely complex local optima; while GNNs are adept at learning the average features of a dataset, they still lack the coordination capability necessary to pursue a global optimum.

5 Experiments and Analysis

5.1 Experimental Setup

We evaluated our architecture on the classic 26-circle packing instance within a unit square. The performance metric is the *Radius Sum* ($\sum r_i$), where a higher value indicates a denser packing. To ensure fairness, all reinforcement learning baselines were trained for 10^5 steps, and all numerical optimization methods were capped at 500 iterations.

5.2 Ablation Study: Efficacy of Components

To verify the contribution of each component in our neuro-symbolic architecture, we conducted an ablation study comparing four configurations:

- **Baseline A (L-BFGS-B Only):** Pure numerical optimization starting from random initialization.
- **Baseline B (PPO + Polishing):** Replacing SAC with Proximal Policy Optimization (PPO), a standard on-policy RL algorithm, followed by L-BFGS-B.
- **Baseline C (SAC Only):** The proposed SAC agent without the deterministic geometric refinement phase.
- **Ours (SAC + L-BFGS-B):** The complete proposed architecture.

The results are summarized in Table 1.

5.3 Analysis of Local Optima and Jamming

The core challenge of circle packing is the prevalence of "jamming states"—rigid local minima where circles are mechanically interlocked.

1. **Pure Numerical Optimization Failure:** As shown in Table 1, Baseline A yields the lowest radius sum. Gradient-based methods like L-BFGS-B are greedy; they rapidly descend into the nearest basin of attraction. Once a contact network is formed (a jamming state), the gradient becomes zero in all feasible directions, preventing topological re-arrangement.

Table 1: Ablation study on 26-circle packing. Success Rate indicates valid overlap-free configurations.

Method	**Best Radius Sum**	**Avg. Radius Sum**	**Success Rate**	**Time (s)**
L-BFGS-B (Random)	2.45	2.38	15%	2.1
PPO + L-BFGS-B	2.51	2.48	65%	80.5
SAC (No Polishing)	2.58	2.55	40%	85.2
SAC+L-BFGS-B	**2.62**	**2.60**	**92%**	88.4

2. **The Role of Maximum Entropy (SAC vs. PPO):** Comparing Baseline B and Ours, SAC significantly outperforms PPO. This validates our hypothesis that the maximum entropy objective ($\mathcal{H}(\pi)$) is crucial. It encourages the agent to maintain stochasticity, effectively preventing it from collapsing into suboptimal jamming states too early. The agent learns to "vibrate" circles out of rigid locks, exploring diverse topologies before the polishing phase locks them in.

3. **The Necessity of Polishing:** While SAC (Baseline C) finds superior global structures, it lacks the numerical precision to reduce overlaps to the 10^{-9} level, resulting in a lower success rate of valid packings. The integration of L-BFGS-B effectively bridges this gap, combining global exploration with local precision.

6 Conclusion

This study presented a neuro-symbolic architecture integrating Soft Actor-Critic with L-BFGS-B for the circle packing problem. Through ablation studies, we demonstrated that the maximum entropy property of SAC effectively alleviates the issue of getting trapped in dense local optima, while the differentiable physics engine and L-BFGS-B polishing ensure geometric validity. Although the result on the 26-circle instance (2.62) does not break world records, the framework provides a robust and scalable methodology for combining learning-based global search with solver-based local refinement. Future work will focus on extending this approach to irregular polygon nesting and 3D bin packing.

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